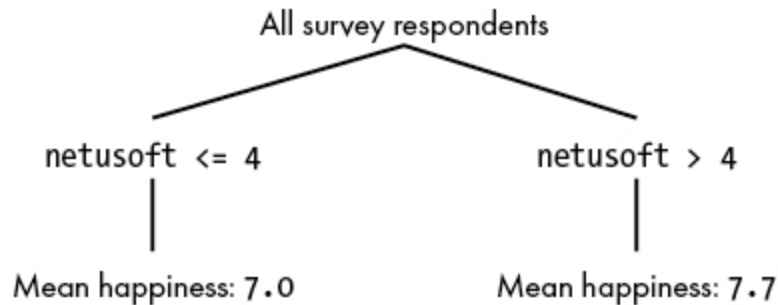


in this branch report their internet usage as 4 or less on a 5-point scale. The last element of each list shows an estimated happiness rating: about 7.0 for those who are not highly active internet users. We can draw a plot of this simple tree in [Figure 9-3](#).



[Figure 9-3](#): The tree generated by our first call to the `getsplit()` function

Our function so far is telling us that people with relatively low internet use report themselves as feeling less happy, with a mean happiness rating of about 7.0, whereas people who report the highest level of internet use report happiness levels at about 7.7 on average. Again, we need to be careful about how we draw conclusions from this single fact: internet use may not be a true driver of happiness, but it may instead be correlated to happiness levels because of its strong correlations with age, wealth, health, education, and other characteristics. Machine learning alone doesn't usually allow us to determine complex causal links with certainty, but, as it has with the simple tree in [Figure 9-3](#), it enables us to make accurate predictions.

Adding Depth

We've completed everything we need to make the best possible split at each branch point and generate a tree with two branches. Next, we need to grow the tree beyond just one branching node and two terminal nodes. Look at [Figure 9-1](#) and notice that it has more than two branches. It has what we call a *depth* of three because there are up to three successive branches you have to follow in order to get the final diagnosis. The final step of our decision tree generation process is to specify a depth that we want to reach, and build new branches until we reach that depth. The way we accomplish this is by making the additions to our `getsplit()` function shown in [Listing 9-4](#).